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INDIA.RUNS

Build what next India runs on

Kisaan Sathi — AI Farm Advisor for India



CHALLENGE 3 — IMPROVE EVERYDAY LIFE WITH AI

TEAM NAME

Agri-Innovators

PROBLEM STATEMENT

Empowering 100M+ Indian farmers by solving the critical expert shortage and income loss due to crop misdiagnosis.

TEAM LEADER

Mihir Aryan Mishra

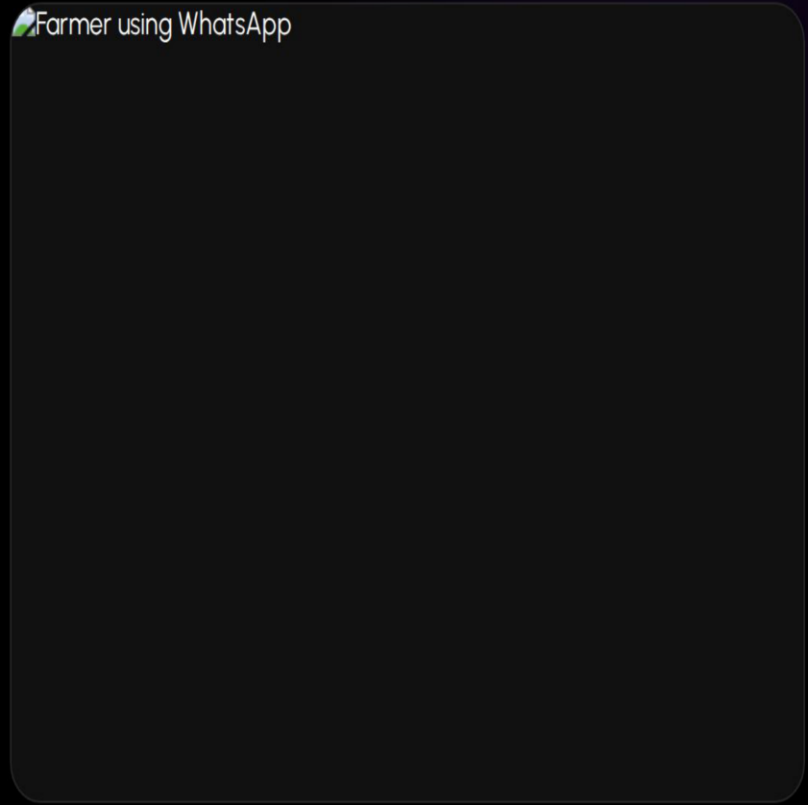
SOLUTION OVERVIEW

What is Kisaan Sathi?

A smart, voice-first AI farm advisor accessible via WhatsApp. It eliminates the literacy barrier and provides expert agricultural guidance in seconds.

- **Voice-First:** Dialect-aware STT models.
- **Zero App Friction:** Works on basic WhatsApp.
- **Multilingual:** 12+ Indian languages.

 Farmer using WhatsApp



USER UNDERSTANDING



Dialect Support

Extracting intent from regional dialects like Bhojpuri and Maithili where standard NLP fails.



Visual Signals

Identifying crop stress and disease patterns from low-resolution photos taken in the field.



Hyper-Local

Real-time data extraction for block-level weather and district-specific Mandi prices.

THE CHALLENGE IN NUMBERS

Expert to Farmer Ratio

1:1000



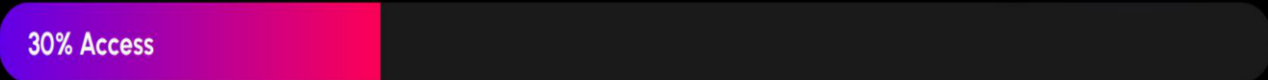
Annual Loss (Cr)

₹6,000 Cr



Scheme Accessibility

30% Access



Kisaan Sathi bridges this gap by providing an "expert in the pocket" for every farmer.

RANKING & ADVICE METHODOLOGY



Data Retrieval

Real-time integration with Agmarknet API and Hyperlocal Weather forecasts.

Scoring Engine

Heuristics-based ranking of treatments based on cost, locality, and crop stage.

Final Decision

Consolidated advice: "Wait 2 days to sell" or "Apply Mancozeb now".



SAFETY & EXPLAINABILITY



Fact Verification

Preventing hallucinations by grounding LLM responses in curated agricultural expert datasets.



Human Escalation

Complexity routing to live Krishi Mitra agents for cases AI flags as uncertain.



Explainable Steps

Every diagnosis includes reasoning: "Yellow spots + high humidity = Bacterial Spot".

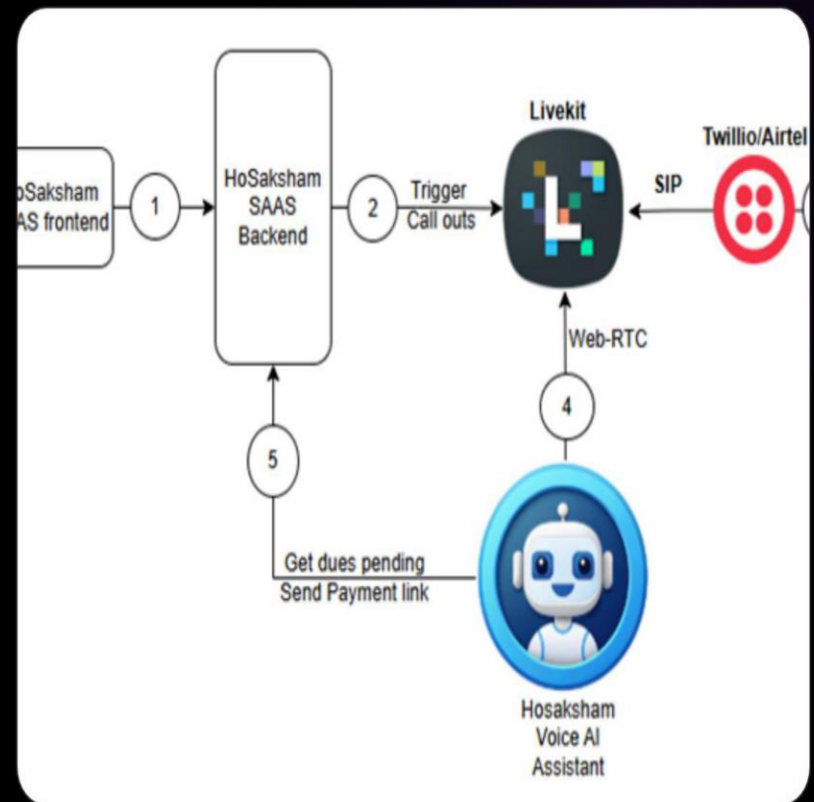
END-TO-END WORKFLOW



SYSTEM ARCHITECTURE

Scalable Tech Stack

- **Edge:** WhatsApp Cloud API for zero-install UI.
- **Compute:** Serverless Python Backend for cost efficiency.
- **Intelligence:** GPT-4o for reasoning + Custom CNNs for crops.
- **Storage:** Vector DB for local agricultural knowledge.



RESULTS & PERFORMANCE

10s

Response Time

- ✓ 30% reduction in crop loss
- ✓ ₹8,000 avg income increase
- ✓ 3x Govt scheme accessibility

SUBMISSION ASSETS



GitHub Repo

Full backend source code
and model training scripts.



Demo Video

End-to-end walkthrough of
a farmer's journey.



Live Prototype

WhatsApp Sandbox link for
interactive testing.

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THANK YOU

Technology Company Presentation Template

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